An La

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Education

2021 - present MS/PhD (Computer Science) at **University of Massachusetts Amherst** (GPA: 3.82/4.0)
Advisor: Professor Hung Le
Selective taken courses: Algorithms with Predictions, Randomized Algorithms, Algorithms for Data Science, Probabilistic Graphical Model, Distributed and Operating Systems.

2013 - 2017 Bachelor's Degree at **Honors Program, VNU-HCMUS, Vietnam** (GPA: 3.57/4.0)
Information Technology, Graduated with distinction.
Final Year Thesis: *Deep Neural Network for Vietnamese News Recognition* (Grade: 10.0/10.0).
Advisor: Professor Vu Hai Quan.

Publications

La, A., & Le, H. (2025, June). *Dynamic Locality Sensitive Orderings in Doubling Metrics*. In STOC'25: Proceedings of the 57th Annual ACM Symposium on Theory of Computing (pp. 1178–1189).

La, A., & Le, H. (2024, August). *New weighted additive spanners*.

La, A., Vo, P., & Vu, T. (2019, July). *Adaptive Collaborative Filtering for Recommender System*. In International Conference on Conceptual Structures (pp. 117-130).

La, A. N. T.*, Nguyen, D. P.*, Pham, N. M., & Vu, Q. H. (2018). *Multi-modal video retrieval using Dilated Pyramidal Residual network*. Science and Technology Development Journal-Natural Sciences, 2(5), 138-143. [1]

Experience

Sep - Dec, 2025: Applied Scientist Intern – Recommender Systems / Information Retrieval at Amazon

- ⋆ Design an explainable exploratory process for time-series forecasting problems.
- ⋆ Design an exploratory assistant for data scientists to investigate statistical models.
- ⋆ Technical skills: *forecasting design*, *prompt engineering*, *LLMs applications*.

July - August, 2025: Data Scientist Intern at GeoComply

- ⋆ Designed and implemented a dynamic clustering framework for robust classification problems.
- ⋆ Designed and developed a multi-agent system for fraud detection.
- ⋆ Improved time efficiency by 4x and discovered new fraud patterns.
- ⋆ Technical skills: *data structure design*, *agentic-system design*, *LLMs applications*.

2021 - now: Research and Teaching Assistant at Theory CS Group/UMass Amherst

- ⋆ Study data structures/algorithms in computational geometry and their applications in data mining.
- ⋆ Study online algorithms with machine learning predictions.
- ⋆ Teaching Assistant: Algorithms for Data Science (Fall22, Spring25), Advanced Algorithms (Spring23).
- ⋆ Technical skills: *algorithm design*, *data structure design*, *algorithmic analysis*.

2020 - 2021: Data Scientist at PrimeData, Vietnam github/anla11/analytic_marketing

- ⋆ Designed and implemented an automatic analysis framework for algorithmic marketing applications.
- ⋆ Technical skills: *quantitative analysis*, *Bayesian machine learning* and *probabilistic programming*.

2017 - 2019: Data Scientist at FPT Telecom, FPT Group, Vietnam github/anla11/adaptive_cf_recsys

- ⋆ Designed and implemented a graph-based model with multiple evaluation metrics for a recommender system.
- ⋆ Increased precision by 6% while maintaining diversity, coverage, and congestion.
- ⋆ Technical skills: *content-based, collaborative-filtering, and graph-based techniques.*

2016 - 2017: Implement image-processing algorithms with standard techniques using OpenCV

- ⋆ Style extraction using GIST and LAB features: github/anla11/style-extraction-for-images
- ⋆ Text region extraction with morphology-based methods: github/anla11/text-region-extract
- ⋆ Image enhancement: github/anla11/image-enhancement

[1] *These authors contributed equally to the work.

Skills

Theory	Design data structures & algorithms, statistics and probability, quantitative analysis and modeling.
Programming	5+ years of implementing projects with Python: implement machine learning models (Tensor Flow, PyTorch, Scikit-Learn); process, analyze and visualize data (Numpy, Pandas, Seaborn). Proficient in C++ to implement algorithms in competitive programming contests and Image Processing (OpenCV).
Other	Git, Latex, Docker, Linux.

Academic Activities

Spring 2025	Industry Mentorship Independent Study - Data Science Supervised master's research projects Publication: Forecasting Time Series with LLMs, to be appeared in PACLIC 39
Spring 2025	Early Research Scholar Programs Supervised undergraduate research projects
June. 2025	Annual ACM Symposium on Theory of Computing (STOC25) present Dynamic Locality Sensitive Orderings in Doubling metrics
Jan. 2025	ACM-SIAM Symposium on Discrete Algorithms (SODA25)
June. 2024	DIMACS Tutorial on Fine-graned Complexity
Jan. 2024	ACM-SIAM Symposium on Discrete Algorithms (SODA24)
Nov. 2022	FOCS 2022 IEEE 63rd Annual Symposium on Foundations of Computer Science
Aug. 2022	FODSI Sublinear algorithms summer school and workshop
June. 2019	Online attending and presenting at 24th International Conference on Conceptual Structures
Aug. 2017	Attending the 3rd Workshop on Statistical Modeling and Applications at VNU-HCMUS Topic: Bayesian Models Inference and Statistical Decision Making

Selective Taken Courses

2023	Distributed and Operating Systems at UMASS Amherst
2022	Algorithms with Predictions, Randomized Algorithms and Algorithms for Data Science at UMASS Amherst
2021	Algorithmic Fairness and Strategic Behavior and Advanced algorithms at UMASS Amherst
2020	Bayesian Methods for Machine Learning - National Research University Higher School of Economics
2019	Probabilistic Graphical Models 1: Representation - Stanford University
2018	Bayesian Statistics: Techniques and Models - University of California, Santa Cruz
2018	Bayesian Statistics: From Concept to Data Analysis - University of California, Santa Cruz
2016	Parallel Programming with GPU, Data Storing and Recovering at VNU-HCMUS

Honors and Awards

Dec. 2016	National Vietnam award for Outstanding Female Students in Science and Technology
Aug. 2016	Awards from Facebook Hackathon Vietnam 2016 *1st prize* of Most Innovative Product *2nd prize* of Best Product in Facebook Marketing Category
2012 - 2014	Vallet Scholarship (South Region) for Excellent Students - https://rvn-vallet.org/
2014	*2nd prize* in ACM-ICPC Vietnam National 1st Round
2013	*3rd prize* in Informatics at the Vietnam National Excellent Student Exam
2012	*Honourable Mention* in Informatics at the National Excellent Student Exam
2011	*Silver Medal* in Informatics at The Traditional 30/4 Olympic Competition